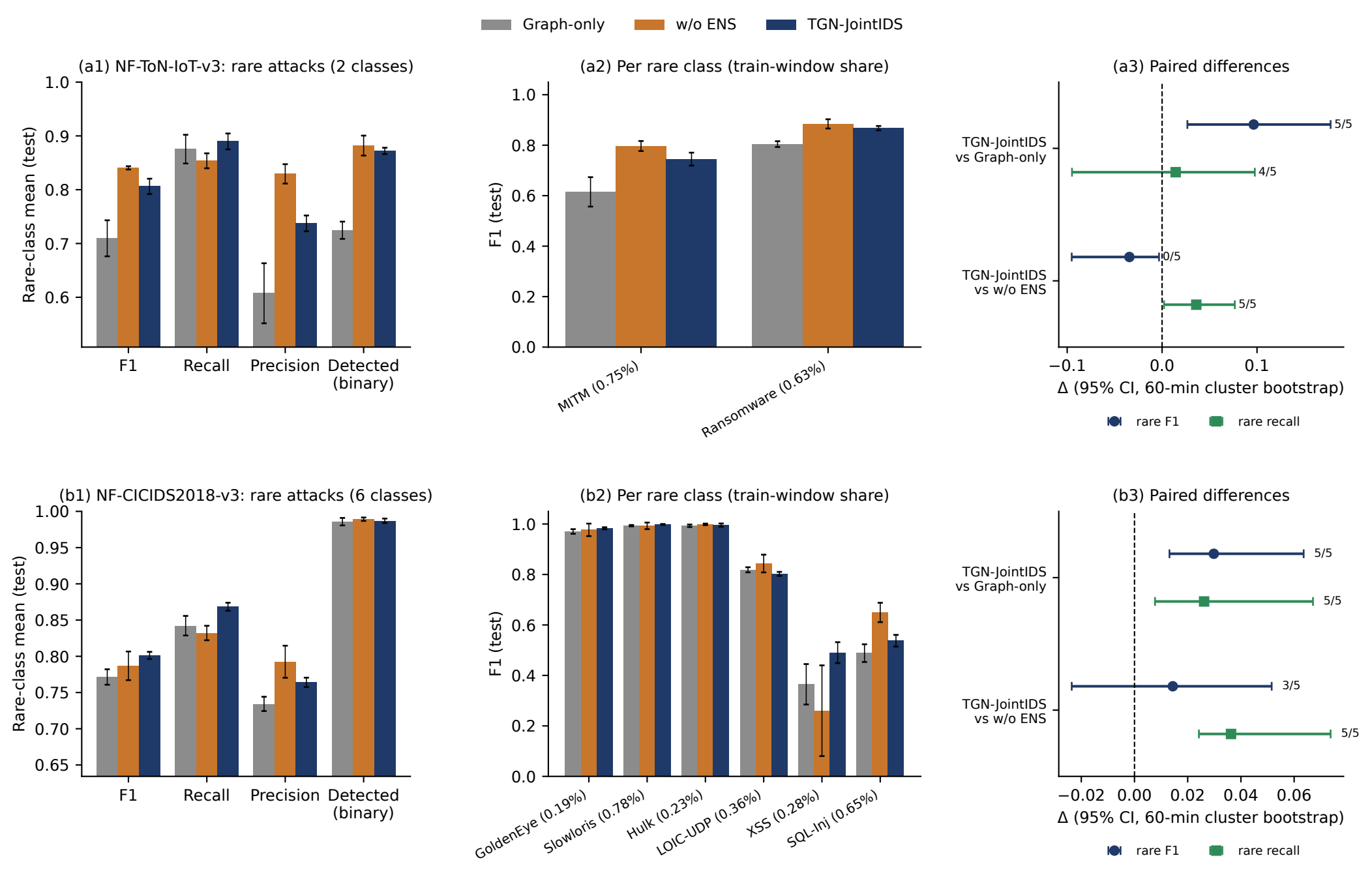
Graph-only
w/o ENS
TGN-JointIDS
(a1) NF-ToN-IoT-v3: rare attacks (2 classes)
Rare-class mean (test)
F1
Recall
Precision
Detected (binary)
(a2) Per rare class (train-window share)
F1 (test)
MITM (0.75%)
Ransomware (0.63%)
(a3) Paired differences
TGN-JointIDS vs Graph-only
TGN-JointIDS vs w/o ENS
5/5
4/5
0/5
5/5
Δ (95% CI, 60-min cluster bootstrap)
rare F1
rare recall
(b1) NF-CICIDS2018-v3: rare attacks (6 classes)
Rare-class mean (test)
F1
Recall
Precision
Detected (binary)
(b2) Per rare class (train-window share)
F1 (test)
GoldenEye (0.19%)
Slowloris (0.78%)
Hulk (0.23%)
LOIC-UDP (0.36%)
XSS (0.28%)
SQL-Inj (0.65%)
(b3) Paired differences
TGN-JointIDS vs Graph-only
TGN-JointIDS vs w/o ENS
5/5
5/5
3/5
5/5
Δ (95% CI, 60-min cluster bootstrap)
rare F1
rare recall